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MECHANICS

(MOTION)

Today

- Motion in a straight line → LINEAR MOTION
- Projectile Motion
  - horizontal projection
  - vertical Projection
  - oblique Projection.

LINEAR MOTION:

straight line.

* This is Motion in a straight line.

Terms Involved in Linear Motion

- Distance
- Displacement
- time
- speed
- velocity
- Average speed
- Average velocity
- Acceleration → Constant OR uniform

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Assume we have (2) fixed points → A, → B

Length between A and B = Distance

↳ S.I unit is metres.

(A) ———————— (B)

RECAP: Distinction between scalar & vector quantities!

Scalar
↓
Have only Magnitude / Size

E.g ↓ time / mass / length → distance

- time → 2s
- mass → 2kg, 120g
- length / distance → 20 Km, 5cm

*Speed is scalar. ($1 ms^{-1}$)

Vector
↓
Have both Magnitude & direction
↳ specific

E.g

* displacement
↓
20Km from A to B

Masaka — Kira

↓ specified direction

* 5cm to the left

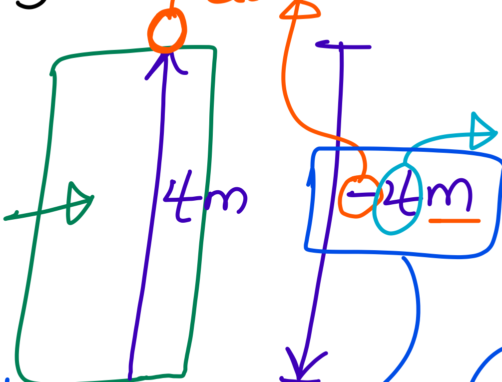
↳ (distance)

↳ specific direction

direction

distance

Building



displacement

*velocity
↳ vector

{ $-5ms^{-1}$, $10ms^{-1}$ (to the left) (at an angle ___)

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Note: You can be given compass directions

→ North, East, West, South

→ N30°W, NE, etc

Time → Measures duration ✓

→ In seconds (s)

→ S.I units.

→ Incase of minutes → 1 min = 60s ✓

1 hr = 60min ✓

1hr = 3600s ✓

→ E.g

✓ time taken to move from A to B.

A → B ✓

✳

time taken to move from A to B

A → B ✓ → t_{BC} ✓ → C ✓

→ t_{AB} → ✓

A combination of t_{AB} ✓ & t_{BC} ✓ is called Total Time Taken (TTT)

Note:

→ distance	d or l
→ displacement	S
→ time	t
→ Total time	T

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Speed → Measured in $\boxed{m/s}$ → ms^{-1}

$m/s \rightarrow \frac{m}{s}$ — m → Measures, s → Measures

$\boxed{\frac{\text{distance}}{\text{time}}}$ = speed

$\boxed{S} = d/t$ → Speed

$t = d/s$ — s → speed

$d = s \times t$ — s → speed

$\boxed{\text{Speed}} = \frac{\text{distance}}{\text{time}}$

distance → change in distance

time → change in time

change in time → $\boxed{\text{RATE}}$

$$X = \frac{(\text{Change in A})}{\text{time}}$$

→ $\frac{\text{Anything}}{\text{time}}$ → RATE

$$\text{Speed} = \frac{\text{distance}}{\text{time}}$$

distance → $\frac{\text{displacement}}{\text{time}}$ → velocity

Speed — ms^{-1} — velocity

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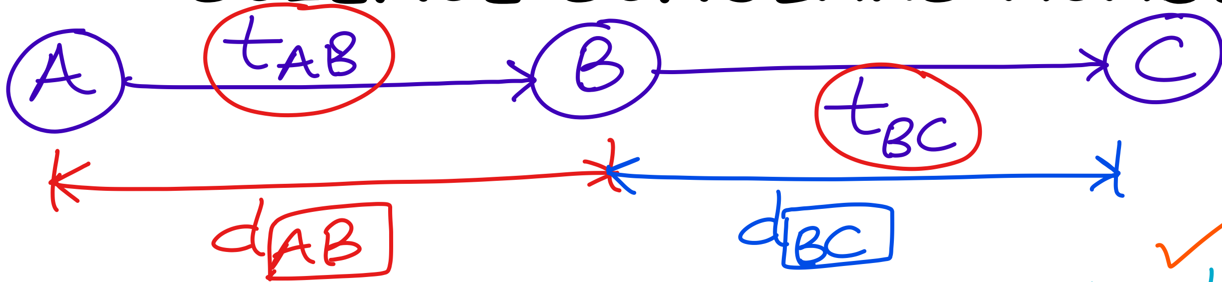


D (total distance) $= d_{AB} + d_{BC}$

T (total time) $= t_{AB} + t_{BC}$

$\frac{\text{total (distance)}}{\text{total (time)}} = \boxed{\text{AVERAGE}}$ Speed

$\frac{\text{total displacement}}{\text{total time}} = \boxed{\text{Average}}$ velocity → vector Qty.

ACCELERATION

→ Measured in $\boxed{ms^{-2}}$

(t) → In Just 5 seconds, velocity increased from $\boxed{4ms^{-1} \text{ to } 20ms^{-1}}$

(A)

$\boxed{ms^{-2}} \rightarrow \frac{ms^{-1}}{s} \rightarrow \frac{\text{velocity}}{\text{time}}$ — change

* velocity changed; from $4ms^{-1}$ to $20ms^{-1}$ in $\boxed{5 \text{ seconds}}$ → (t)

Initial velocity (u) final velocity (v)

(*) Acceleration is denoted by $\underline{\underline{a}}$.

But Acceleration = $\dfrac{\text{Change in velocity}}{\text{time}}$

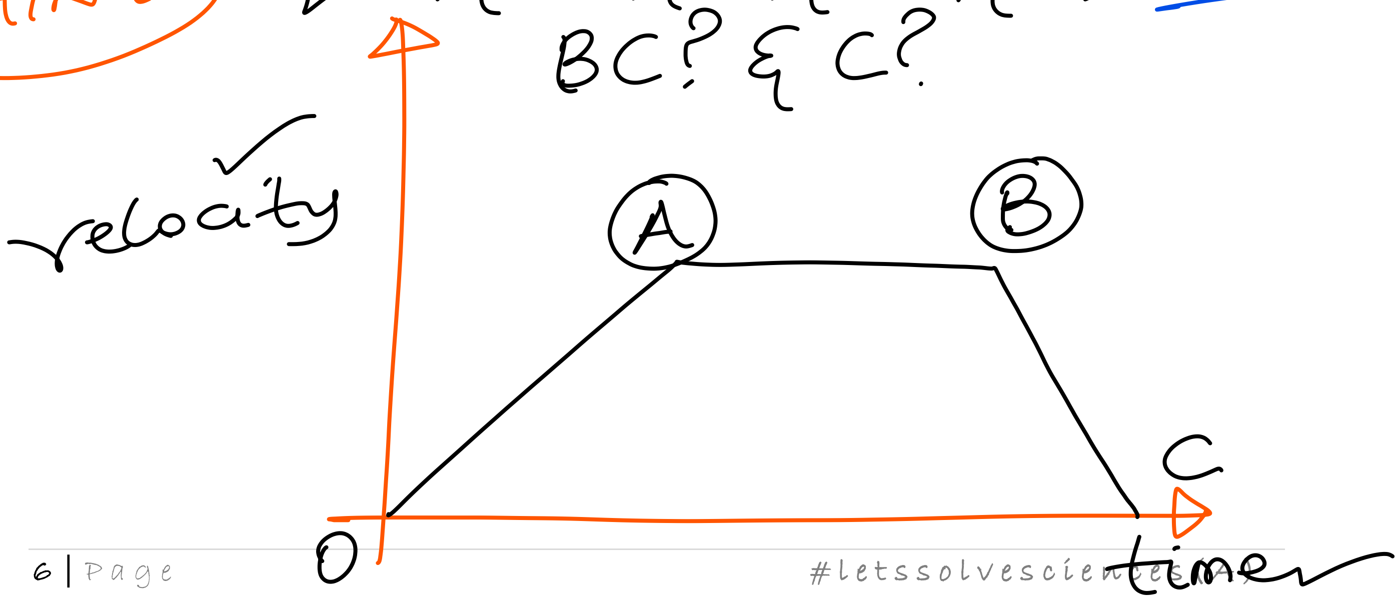
$$\boxed{a = \frac{v - u}{t}} \rightarrow (*) ✓$$

(*) If a body has both u (Initial velocity) & v (final velocity)

Average (Mean ✓) velocity $= \left(\dfrac{u+v}{2}\right)$ ✓

Research: Sketch a well-labelled velocity-time graph. → UNEB 2024 P2.

HINT → O? / OA? / A? / AB? / B? BC? & C?



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3 EQUATIONS OF MOTION.

from; $\boxed{a} = \frac{v-u}{t}$

$at = v - u$

$$\boxed{v = u + at} \longrightarrow \text{1st Eqn of motion!}$$

from; Average velocity = $\frac{\text{total displacement} \rightarrow S}{\text{time} \rightarrow t}$

$\left(\frac{u+v}{2}\right)$

$$\left(\frac{u+v}{2}\right) = \frac{S}{t}$$

$$S = \left(\frac{u+v}{2}\right)t \longrightarrow \boxed{***}$$

from Eqn ①, $v = u + at$

$$S = \left[\frac{u + (u+at)}{2}\right]t$$

$$S = \left(\frac{2u+at}{2}\right)t$$

$$S = \frac{2ut + at^2}{2}$$

Recall

$$\left\{ \frac{x+y}{z} = \frac{x}{z} + \frac{y}{z} \right\}$$

$$S = \frac{2ut}{2} + \frac{at^2}{2}$$

$$S = ut + \frac{1}{2}at^2$$ Eqn (2) of motion.

(3). from Eqn (1); $a = \frac{v-u}{t}$

$$\boxed{t = \frac{v-u}{a}} \rightarrow (**)$$

Substitute (**) into (***)

$$S = \left(\frac{v+u}{2}\right)\left(\frac{v-u}{a}\right)$$

$$S = \frac{(v+u)(v-u)}{2a}$$ Research.

$$S = \frac{v^2 - u^2}{2a}$$

$$2aS = v^2 - u^2$$

$$\boxed{v}^2 = \boxed{u}^2 + 2\boxed{a}S$$ Eqn (3) of Motion.

✓✓ ✓✓ ✓✓ ✓✓

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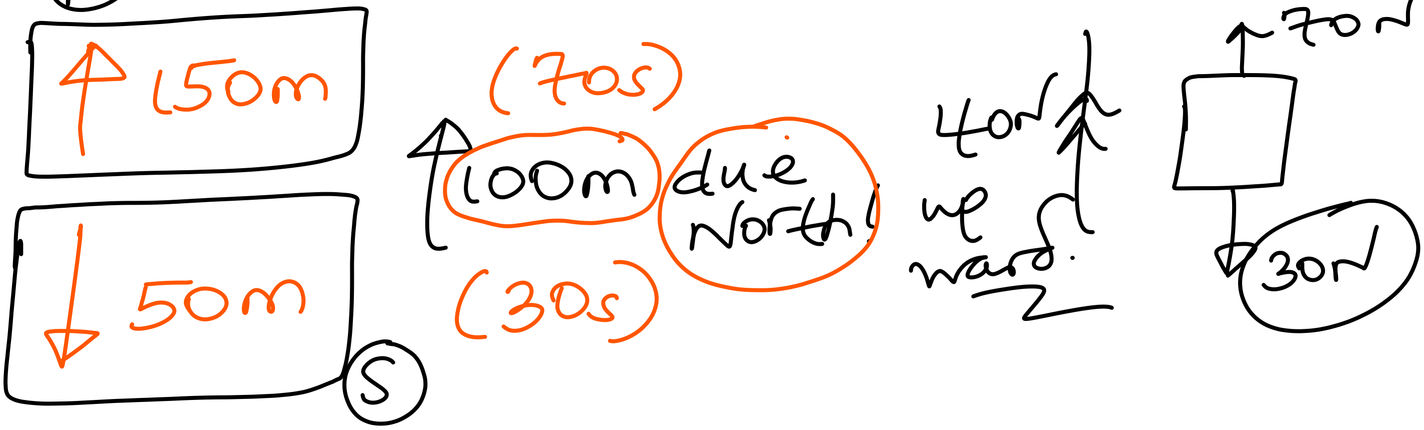
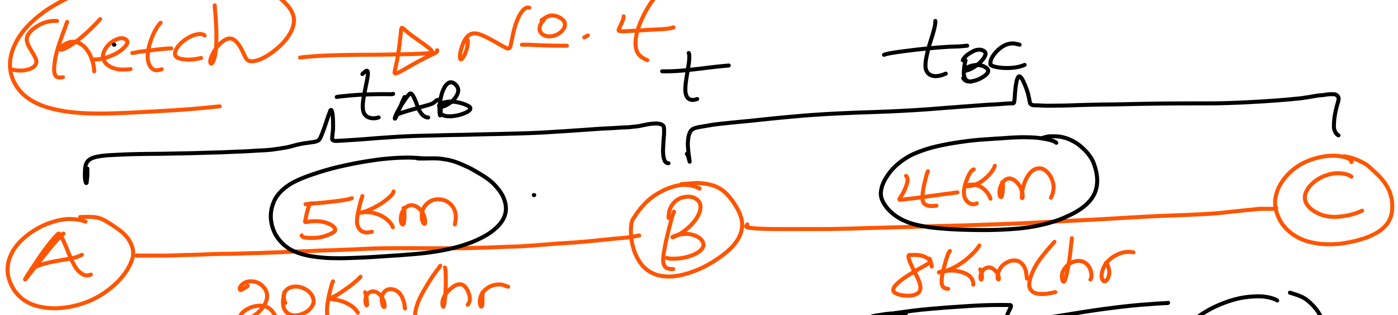


Sketch → No. 4



(a). $t_{AB} = 5/20 = \frac{1}{4}\,hr$ $\qquad t_{BC} = 4/8 = \frac{1}{2}\,hr$

$$T = \frac{1}{4} + \frac{1}{2} = \frac{3}{4}\,hrs$$

$(3/4 \times 60)$ minutes

$(3/4 \times 60 \times 60)$ seconds

1. ✳ u $(t$ $(a$ $(s?$ $\quad u \to 0,\ t \to 2,\ a \to 4$

$$s = ut + \frac{1}{2}at^2$$

Note: Eqns of Motion are in ē Log book → Pg. 65 (formula 77)

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Solution: $u = 0\,ms^{-1}$; $t = 2s$; $a = 4\,ms^{-2}$

(i). $S = ut + \frac{1}{2}at^2$

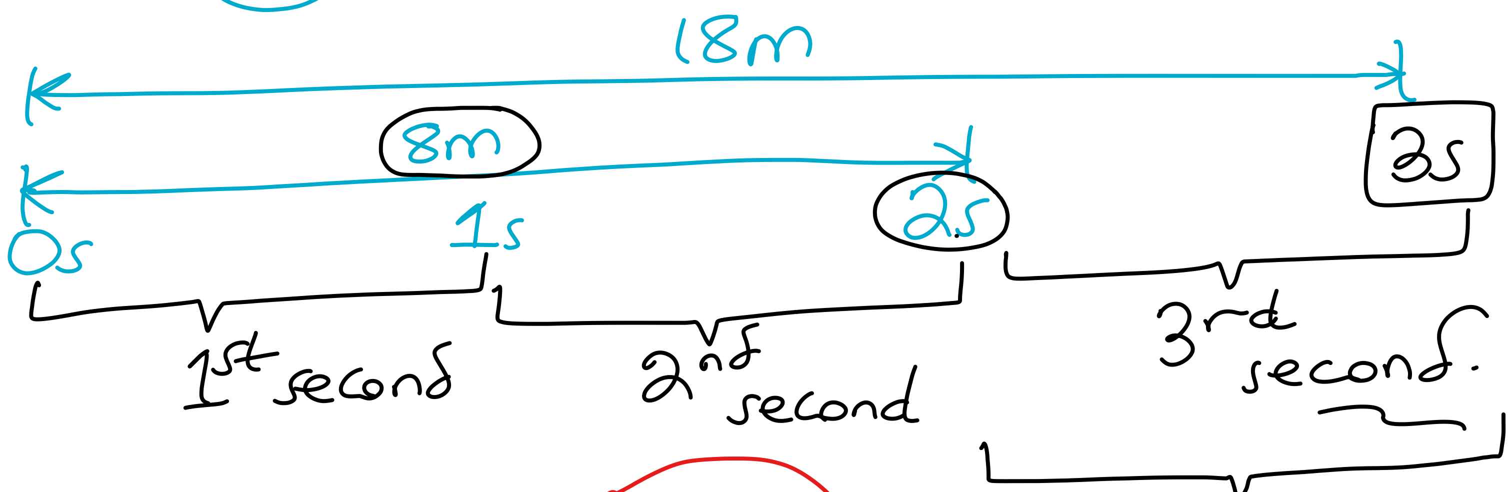
$S = 0(2) + \frac{1}{2}(4)(2^2)$

$S = 8m$

(ii). $t = 3s$

$S = 0(3) + \frac{1}{2}(4)(3)^2$

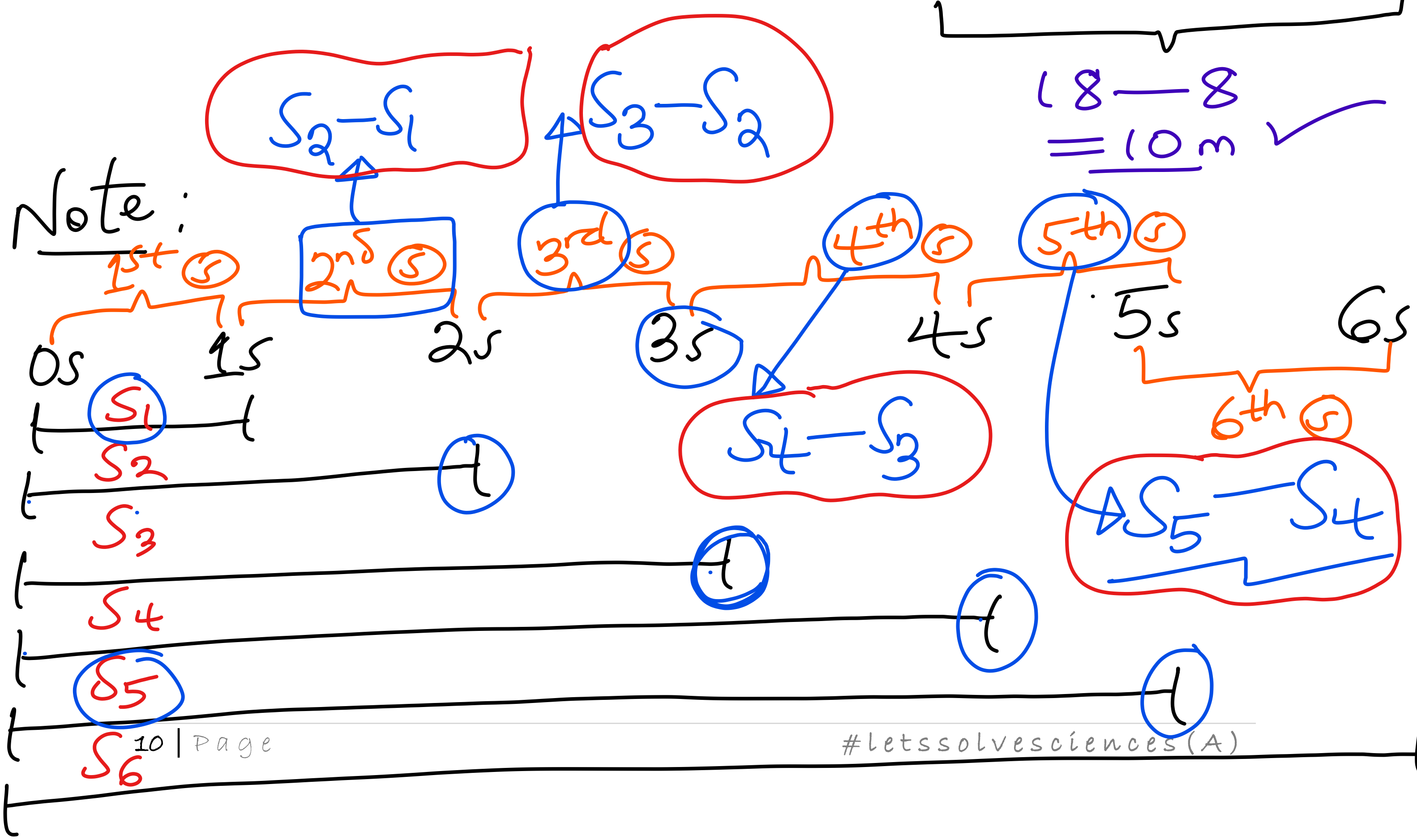
$S = 18m$



$18 - 8$

$= 10\,m$

Note:



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$$S = u\boxed{t} + \frac{1}{2}a\boxed{t}^2$$

S in 100th second

$$S_{100} - S_{99}$$

* S in the 96th second

$$\boxed{S_{96}} - \boxed{S_{\boxed{95}}}$$

$$S = ut + \frac{1}{2}at^2 \leftarrow S = u\boxed{t} + \frac{1}{2}a\boxed{t}^2$$

Acceleration (+) vs deceleration (−)

Positive acc. — Negative acc

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$$v^2 = u^2 + 2aS$$

$$\frac{\boxed{v}^2 - \boxed{u}^2}{2\boxed{a}} = \boxed{S}$$

$v \to 0$, $u \to 10$, $a \to -2$, $S \to 25$

$$t = \frac{v - u}{a}$$

t, v, u, a

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